Code No: 114CS **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, August/September - 2022
DESIGN AND ANALYSIS OF ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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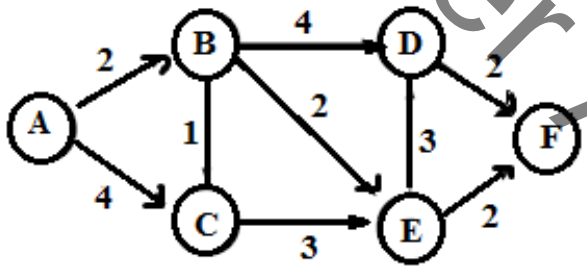
1.a) Explain the Binary Search algorithm and derive the time complexity.
b) Explain about the different types of asymptotic notations. [6+9]

2.a) Derive the time complexity of Strassen's matrix multiplication.
b) Explain about the Quick sort with an example. [7+8]

3.a) Write an algorithm of AND/OR graph and also analyze the time complexity of the same.
b) Explain how to improve the union algorithm with an example. [8+7]

4.a) Write a non recursive algorithm of preorder tree traversal.
b) Explain the Bi-connected components. [8+7]

5.a) Find the shortest path from source(A) to destination (F) by using single source shortest path problem.



b) Differentiate between Greedy and Dynamic Programming. [8+7]

6.a) Write an algorithm of Prim's minimum cost spanning tree.
b) Explain the Reliability design problem. [8+7]

7.a) w = {15, 7, 20, 5, 18, 10, 12}, m = 35. Find all possible subsets of w that sum to m. Do this using sum of subsets. Draw the portion of the state space tree that is generated.
b) Write an algorithm of FIFO Branch and Bound. [8+7]

8.a) Write a Non-deterministic algorithm of 0/1 Knapsack problem.
b) Explain about the Cook's theorem. [8+7]

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